QP Code

Reg.No:
Name:

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION MONTH 2025
THIRD SEMESTER
DSC-MG3DSCBIF201-Applied Mathematics
(2024 ADMISSION ONWARDS

Duration: 2 hrs **Maximum Marks: 70 marks**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Define a subset with example [U]/ [CO1]

2. Define null matrix, give an example [U] / [CO7]

3. Construct the contrapositive of the implication $p \rightarrow q$ [A] / [CO3]

4. Translate the statement into logical form [A] / [CO4]
" *If it rains, then the ground gets wet*"

5. Define degree of a vertex [K] / [CO5]

6. What is a sub graph [U] / [CO6]

7. Calculate mean of the following data 12, 15, 18, 20, 25 [Ap] / [CO8]

[2x5 = 10]

Part B

Short Essay Questions

Answer any**5** questions

Each question carries **6** marks

8. Let $A = \{1, 2, 3, 4, 5\}$ $B = \{3, 4, 5, 6, 7\}$Find: [An] / [CO2]
a) $A \cup B$
b) $A \cap B$
c) $A - B$
d) $B - A$

9. Prove $AB \neq BA$ where $A = \begin{pmatrix} 3 & 2 & -8 \\ 5 & 1 & 0 \\ -1 & 2 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 & 2 \\ 5 & 4 & -1 \\ -2 & 4 & 3 \end{pmatrix}$ [U] / [CO7]

10. Use De Morgan's Laws to simplify the logical expression: $\neg(P \wedge Q) \vee (\neg P \wedge Q)$ [A] / [CO3]

11. Construct the truth table for the formula: $(P \rightarrow Q) \wedge (\neg Q \rightarrow \neg P)$ [A] / [CO4]

12. State and explain Handshaking Theorem with an example. [K] / [CO5]

13. Calculate the mean, median, and mode for the following data: 12, 15, 14, 16, 15, 18, 15, 17, 13; Interpret the result [Ap] / [CO8]

14. Differentiate between a valid argument and an invalid argument with examples [CO3]

15. Explain the double negation law with one example from real life [CO3]

[6x5 = 30]

Part C

Long Essay Questions

Answer any **3** questions

Each question carries **10** marks

16. Solve the following system using matrices: [U] /[CO7]

,$2x + y = 5$

$$3x - 2y = 4$$

17. Explain the Types of Degrees and Their Properties in Graphs [U] / [CO6]

18. Using a truth table, determine whether the following compound proposition is a tautology, contradiction, or contingency: $(p \rightarrow q) \leftrightarrow (\neg p \vee q)$ [A] / [CO5]

19. Describe the various types of graphs and diagrams used in biostatistics to [Ap] / [CO8]

represent data. Explain with suitable examples.

20. What is meant by logic in mathematics. Why is it considered important in solving mathematical problems? [CO3]

[10x3 = 30]